

Kona Biopsy Age Ranking Models: Comparison Brief

Prepared from attached Models 1-3 workbooks. No database records or raw data were modified for this comparison.

Bottom Line

The age-ranking system is a pairwise evidence model, not an absolute age estimate. A sample ranks older when there is dated evidence that its manta was mature before another sampled manta was documented as immature. Tied ranks are expected when the evidence cannot distinguish samples.

The major shift occurred from Model 1 to Model 2: 11 samples moved older/higher, 71 moved younger/lower, and 1 retained the same rank. Model 3 did not change any ranks or scores relative to Model 2 because Model 2 was already treating pup labels as immature; Model 3 made the pup-as-birth-year assumption explicit and added audit columns.

What Each Model Includes

Each model keeps the same 83 biopsy samples. The difference is not the sample set; it is which dated maturity/immaturity evidence is allowed to create pairwise older-than comparisons. The models are nested: Model 2 includes everything in Model 1, and Model 3 includes everything in Model 2.

Model	Explicitly includes	New vs prior	Still excludes / caveat
Model 1	HAMER biopsy-linked age class, which may already include size-based maturity/immaturity decisions, plus MPRF >15-year resight history counted as mature at first sighting +15 years.	Starting model.	Does not use MPRF adult/juvenile/pup labels, pup-as-birth-year, independent raw-size modeling, notes, photos, or sex-specific evidence unless already embedded in the HAMER age class.
Model 2	Everything in Model 1, plus MPRF-provided age classes. Adult/mature is treated as mature; juvenile/pup/immature is treated as immature on the MPRF first/effective sighting date.	Adds medium-confidence MPRF age-class labels, increasing pairwise edges from 43 to 597.	Still does not separate pup from juvenile as a birth-year anchor, and does not independently verify the field criteria behind MPRF age classes.
Model 3	Everything in Model 2, plus explicit pup birth-year anchors and immature/pup +15-year mature-date fields when later sighting history supports that derived date.	Adds transparency/audit fields for pup and +15-year logic; no rank or score changes in the current files.	Still does not independently re-evaluate raw size measurements, measurement trends, reproductive notes, clasper notes, photos, or unresolved QC corrections.

Size treatment: raw size measurements were not re-modeled independently in any of these three models. However, HAMER age-class calls may already include size-based maturity or immaturity decisions made in the field/database, so that size-informed evidence is incorporated through the HAMER age-class field in Model 1 and carried forward into Models 2 and 3.

Evidence density increases from 43 pairwise edges in Model 1 to 597 in Model 2. Model 3 remains at 597 edges because, in these files, the new pup-birth-year fields clarify the evidence rather than adding new rank-separating comparisons.

How Rankings Changed

Transition	Moved older	Moved younger	Same rank	Score changed
Model 1 -> 2	11	71	1	75
Model 2 -> 3	0	0	83	0

Interpretation: Model 2 supplies many more immature anchors through MPRF age classes, increasing the number of pairwise comparisons and spreading the ranking. Older/mature individuals tend to move higher because they are now ordered before many more MPRF juvenile/pup samples. Samples with only immature evidence tend to move lower because they are now explicitly younger than more mature-dated samples.

Largest Model 1 to Model 2 Rank Movements

Sample	Source	M1	M2	Delta	M2 evidence
Melainah MP147 Cat59	HAMER	75	39	36	immature
Not RR F Cat71	HAMER	75	39	36	immature
Tucker MP206 Cat68	HAMER	75	39	36	immature
Vallaray MP110	HAMER	75	39	36	immature
McBoom	MPRF	9	83	-74	immature
Bob Smith	MPRF	9	80	-71	immature
Chloe	MPRF	9	80	-71	immature
Kamala	MPRF	9	80	-71	immature

Confidence Context

- Model 1 is the cleanest parent/offspring exclusion screen because it relies on HAMER field maturity calls, which may already incorporate HAMER size thresholds, and long MPRF resight histories rather than broad MPRF age-class assumptions.
- Model 2 is useful for ranking all 83 samples but lower confidence for MPRF-only age classes, because the underlying maturity criteria may not be consistently documented.
- Model 3 is not a new rank outcome in the current files; its value is transparency. It preserves the pup birth-year anchor and derived +15-year maturity dates for review.
- All models remain sensitive to ongoing QC of sighting, manta, photo, biopsy, and catalog links. Treat ranks as relative evidence weights, not definitive ages.

Recommended Next Checks

- Review the 32 pup birth-year anchors and confirm whether each label is defensible as a near-birth-year observation.
- For MPRF adult/juvenile labels, inspect notes, photos, size records, mating scars, pregnancy, mating-train context, and clasper evidence where available.
- Add independent raw-size modeling only after checking measurement consistency over time; large up/down size fluctuations should be flagged before use.
- Use the master CSV as the audit table: it shows every sample, all model ranks/scores, evidence basis, movement direction, and Model 3 pup/+15-year fields.

Source: attached Model 1, Model 2, and Model 3 workbooks in the Kona Biopsy Age Rankings folder.